

Module 12: Free Space Management Techniques

Unit 5: I/O & File Systems

AKTU BCS401

Disk Management

1. Concept & Architectural Goals

Disk par files continuously create, modify aur delete hoti rehti hain. Operating System ko ye track rakhna padta hai ki kaun se blocks free hain aur kaun se allocated. Primary goals: **Fast Allocation**, **Minimal Memory Overhead**, and **Support for Contiguous Blocks**.



2. Core Free Space Management Techniques

2.1 Bit Vector / Bitmap

Har disk block ke liye 1 bit reserve hoti hai (0 = Allocated, 1 = Free). First free block calculate karne ka standard CPU hardware formula:

$$\text{Block Number} = (\text{Number of 0-words} * 32) + \text{Offset of First 1-Bit}$$

Modern CPUs me hardware instructions jaise bsf (bit scan forward) aur ffs (find first set) O(1) time me bit detect karte hain.

Bitmap RAM Overhead Derivation:

Agar Disk Size = 1.3 TB, Block Size = 512 B:

Total Blocks = $1.3 \times 10^{12} / 512 = 2,539,062,500$ blocks.

Bitmap Size = $2,539,062,500 \text{ bits} / 8 = \mathbf{302.68 \text{ MB RAM}}$.

Agar Block Size 4 KB kar dein: Bitmap Size = **37.8 MB** (8x memory savings!).

2.2 Linked Free List vs Grouping vs Counting

Parameter	Bit Vector	Linked List	Grouping	Counting
RAM Overhead	High (Entire bitmap in RAM)	Minimal (Head pointer)	Low (Current group block)	Low (B-Tree nodes)
Disk Overhead	Dedicated bitmap space	Zero (Stored inside free blocks)	Zero (Stored inside free blocks)	Small (Extent table)
Contiguous Search	Easy (Bit shift / AND masks)	Disastrous (High seek count)	Poor	Optimal (Native)
Modern Use	ext2/3, NTFS	Rare / Legacy	UNIX System V	ext4, XFS, ZFS

Grouping: Single free block stores (N-1) free block addresses + 1 pointer to next group block. 1 disk read provides hundreds of free block locations.

Counting: Stores tuples (start_block, count) for contiguous extents. Drastically compresses allocation metadata.